

continuation

Continuation Document

Revision: 67 **Last modified:** 2026-06-01T20:57:15Z **Description:** Sacred invariant resumption document for Helix Cluster OS **Authority:** Constitution §12.10
Maintainer: Operator + AI loop

§0: How to use this document

Any CLI agent resuming work on this project MUST read this file first.

§1: Recently Completed Work (last 10 commits)

Commit	Message
888a5a3	Foundation wave 15: mark 4 items Completed (HXC-1136/1137/1231/1260) — session-create EXIT GATE proven vs real tmux / scheduler-latency EXIT GATE <100ms / provisioner-lifecycle-FSM / KPI-quality-gate. MVP exit-gate triad complete (1135+1136+1137). Registry 161->165 Completed / 474 Queued.
553a043	Foundation wave 15: 4 streams, 4 items incl 2 MVP exit gates. HXC-1136 session-create EXIT GATE — wired pkg/session.Manager to a REAL tmux/PTY backend via BackendAdapter (fixed the nil-backend StatusRunning PASS-bluff in internal/session.NewServer; added NewServerWithTmuxBackend seam + SessionExists sink-probe); E2E proves htmux create -> helix-session -> real tmux session visible running in 388ms (<3s). HXC-1137 scheduler decision-latency EXIT GATE (<100ms across 10 jobs, histogram + placement-correctness + 120ms-slow-plugin mutation). NEW pkg/testing/instance Provisioner/Instance lifecycle FSM (provisioned->booting->ready->stopped on virtual clock, illegal-transition reject). NEW pkg/qualitygate KPI baseline validation + critical-severity CI gate (8 rules). ZERO new deps. GATE caught+fixed a real regression: 1136's real backend exposed test/chaos+test/e2e relying on the no-op bluff (1000 free sessions -> 509 leaked tmux -> PTY exhaustion) -> switched those harnesses to the in-memory NewServerWithTmuxBackend(nil) seam + added SessionExists to the test double. Full -short -race green, 1136 real-tmux E2E + 1231/1260 integration green, live mutation spot-check on qualitygate critical gate confirmed real.
34efa5d	Foundation wave 14: mark 4 items Completed (HXC-1135/1229/1247/1261) — MVP two-node-formation EXIT GATE proven vs real etcd / chaos-sequencer+Byzantine / device-profile-registry / Welch-t-test. Registry 157->161 Completed / 478 Queued.

Commit	Message
1c0d943	Foundation wave 14: 4 disjoint streams, 4 items incl the MVP two-node-formation EXIT GATE. cmd/helix-agent -etcd-endpoints/HELIX_ETCD_ENDPOINTS wiring + HXC-1135 E2E (two REAL helix-agent processes form a cluster via real etcd, both visible <60s w/ per-run UUID + formation timestamp + evidence file); pkg/testing/chaos seeded deterministic fault sequencer (reproducible injection log) + Byzantine equivocation fault; NEW pkg/deviceprofile versioned-YAML device profile registry T1-T8 + schema validation/rejection; NEW pkg/stats Welch t-test regression detector (t-stat/Welch-Satterthwaite df/p-value). ZERO new deps. GATE: build/vet/vet-integration clean, full -short -race green, 1135 two-process formation PROVEN vs REAL etcd (2.76s), deviceprofile+stats integration green, live mutation spot-check on Welch significance confirmed real.
e1dab02	Foundation wave 13: mark 6 items Completed (HXC-1189/1190/1198/1210/1213/1220) — gateway-auth-deny-reason / node-etcd-wiring (real etcd cross-visibility) / OPA-decision-log / EdgeAware-Filter+Score / offline-sync-delta-compression. Registry 151->157 Completed / 482 Queued.
203c42a	Foundation wave 13: 5 disjoint-package streams, 6 items. internal/gateway X-Helix-Deny-Reason header + scheduler:write scope enforcement (401/403 reject-before-proxy); internal/node REAL EtcdBackend wiring (Config.EtcdEndpoints -> NewEtcdBackend w/ SWIM-tied TTL lease + resource metadata, BackendType() status, in-memory fallback) — node registrations now cluster-visible (unblocks 1135 two-node formation); internal/policy real persisted OPA decision-log sink (DecisionLogEntry RunUUID/Allow/Reason/PolicyName); NEW pkg/scheduler/edgeaware.go EdgeAware Filter (tier>=6 reject session/sleeping/over-thermal/not-charging w/ reasons) + Score (per-tier+thermal-headroom bonuses); NEW pkg/offlinesync delta-compression (flate) offline job-ledger + idempotent reconcile (100% recovery). ZERO new deps. GATE: build/vet/vet-integration clean, full -short -race green, integration tier 1210 vs REAL etcd (A sees B+metadata, UUID) + gateway/policy green, live mutation spot-check on edgeaware T6 guard confirmed real.
23df87b	Foundation wave 12: mark 14 items Completed (HXC-1148/1149/1150/1151/1156/1157/1188/1191/1195/1196/1211/1212/1214/1215) — 12 integration-proven vs real etcd/minio + 2 gate-verified already-implemented (STUN/OTLP). Registry 137->151 Completed / 488 Queued.
7718c20	Foundation wave 12: 5 disjoint-package streams, 12 items integration-proven vs real etcd/minio. pkg/wireguard typed ErrUnsupported (errors.Is) + key-rotation grace/overlap ActiveKeys() consumed by mesh; pkg/scheduler named ClassAdFilter/ScorePlugin (classads Requirements/Rank) + thermal-aware plugin (filter/score on temp_c threshold); internal/scheduler real PREEMPTED lifecycle event (terminal, ordered); NEW pkg/edge trust model (FULL/STANDARD/SEMI/EDGE_DONOR) + workload restriction matrix + work-unit resource-limit executor (kill-after duration) + declarative per-tier ScheduleRule engine; NEW internal/verifier SEMI output verification (sha256 redundant-recompute reject tamper) + pkg/discovery etcd TTL-lifecycle integration test + pkg/storage real-minio S3 round-trip integration test. ZERO new deps. GATE: build/vet/vet-integration clean, full -short -race green, integration tier vs real etcd(:38623 ephemeral)+minio(:9000) green, mutation spot-check on verifier confirmed real.
6715b3a	Foundation wave 11: mark 3 items Completed (HXC-1122/1123/1124) — integration-proven vs real capnp/clang++/libzmq
63822b3	Foundation wave 11: Cap'n Proto zero-copy serde (node.capnp + capnpc-go, AllocsPerRun==0), FlatBuffers GPU compute payloads (compute.fbs + flatc Go + pooled builder + clang++ C++ consumer harness), ZeroMQ data-plane (ROUTER/

Commit	Message
	DEALER correlated + PUSH/PULL at-least-once, pebbe/zmq4) — integration-proven (real capnp/clang++/libzmq). Toolchains capnp/flatc/libzmq + Go runtimes wired. GATE: fixed C++ harness flatc root-accessor (GetRootAsComputeTask->GetComputeTask)

§2: Environment Snapshot

Property	Value
Branch	main
Commit	888a5a3
Timestamp	2026-06-01T20:57:15Z

§3: Active Work

HXC	Title	Status
HXC-922	Node service, advisory locks, chaos tests, docs	✓ Done
HXC-921	Web UI, K8s/Helm, integration, E2E, benchmarks	✓ Done
HXC-920	Health, security, build, wireguard services	✓ Done
HXC-919	htmux, messaging, GPU, WASM, storage, build manifest	✓ Done
HXC-918	Wire scheduler/session gRPC to real backends, etcd discovery	✓ Done
HXC-916	Stub expansion (config, retry, ratelimit, validator, build cache)	✓ Done
HXC-914	LLM, policy, setup, distributed lock	✓ Done
HXC-913	Health, security, ClassAds parser	✓ Done
HXC-912	Gateway, helixd, helix-agent	✓ Done
HXC-911	Scheduler, Session gRPC services	✓ Done
HXC-910	Console, security, testing infra	✓ Done

§4: Next Planned Work

All MVP, Phase 2, Phase 3, and Phase 4 are **COMPLETE**. Potential future enhancements:

1. **Performance optimization** — Profile and optimize hot paths
2. **Multi-region support** — Cross-region cluster federation
3. **GPU scheduling v2** — Multi-GPU, fractional GPU allocation
4. **Web UI real-time** — WebSocket integration for live updates
5. **Operator pattern** — Kubernetes operator for cluster management

§5: Known Issues / Blockers

- None

§6: Quick Commands

Run all tests (requires GOWORK due to workspace modules)

```
GOWORK=$(pwd)/go.work go test ./... -race -count=1
```

Build all binaries

```
GOWORK=$(pwd)/go.work go build ./cmd/...
```

Build web UI

```
cd web && npm run build
```

Render Helm chart

```
helm template helix-cluster deploy/helm/
```

Run benchmarks

```
go test -bench=. ./test/benchmark
```

Run chaos tests

```
go test -race -count=1 ./test/chaos
```